

Transition anatomy at n_{stab}

TRANSITION_SMOOTH: no universal cliff; the Schur plateau is the stable prefix

Stefan Hamann

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Abstract

The Prefix Stability Theorem candidate needs a property that is stable on the early chain and may be lost later. This note compares three single-window objects, profiled in rank n and centred on the r449 cut n_{stab} : the OP conditioning/Christoffel regime; fold-atom Nyquist n_{res} ; and the compressed dual-resolvent energy $q_n^\dagger$. None of the three kinks universally at n_{stab} (TRANSITION_SMOOTH). Atom Nyquist is $L/2$ and $n_{\text{stab}}/n_{\text{res}} \in [0.010, 0.094]$ (RES_MISMATCH): the prefix is bandlimited. What *is* stable on $n \leq n_{\text{stab}}$ of every measured window is a constant Schur value $q_n^\dagger \equiv q_* < 1$ (PREFIX_Q_PLATEAU). n_{stab} is a sufficient operational cut for that plateau, not a unique mechanical cliff. No RH claim. No anti-RH claim.

Status. No new SATZ. Propositions 2–4 are a named infra census. Ausgang TRANSITION_SMOOTH / RES_MISMATCH / PREFIX_Q_PLATEAU. No L^* claim. No $R^\dagger$ claim. No RH claim. No anti-RH claim.

1 Objects

n_{stab} is the r449 Fenster-Vergleich (Chebyshev moments of folded $\tilde{\mu}$, skip m_0, m_1 , first neighbour break; the cut is identical from $\varepsilon = 10^{-6}$ to 10^{-2}). It is *not* redefined here.

Definition 1 (n_{res} , source-pure). *Let $\{x_i\}$ be the folded μ -atoms of the window, $x_i \in [-1, 1]$. Set $\theta_i = \arccos(x_i)$ and*

$$n_{\text{res}} = \pi / \min_i (\theta_{i+1} - \theta_i)$$

after sorting unique angles. No fit degree of freedom. On the cosine grid of the fold this equals $L/2$.

Definition 2 ($q_n^\dagger$). *The r362/r369 pack cut at depth n : μ -chain plus smooth-border augmentation, $B_w = \text{cumsum}(\rho)[n - 2] + 5/7$, $q^\dagger = \|\cdot\|$, $\delta = 1 - q^\dagger$. n_{qbreak} is the first $n > n_{\text{ref}}$ with $|q_n - q_{\text{ref}}| > 10^{-4}$.*

2 Conditioning

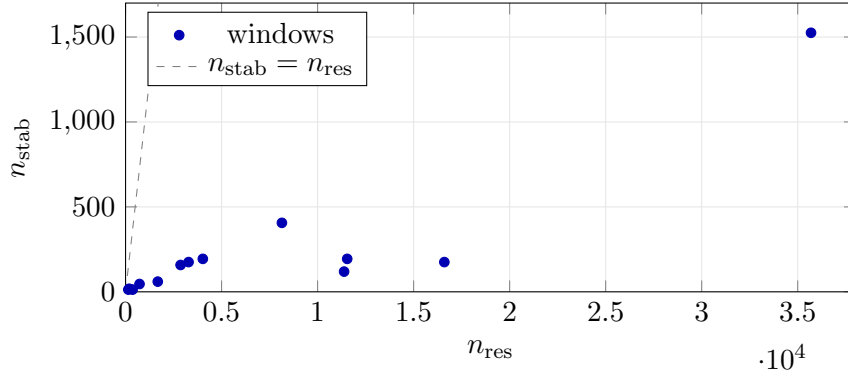
Proposition 1 (No Christoffel cliff). *On kz17, 136, 137, 197, the Jacobi couplings stay $\beta_n \approx 1/2$ through n_{stab} and $n \lambda_n(0) \in [2.3, 6.9]$ (continuous regime). A slope-break of $|\beta_n - 1/2|$ is not peaked within 15% of n_{stab} with ratio > 4 . The discrete near-singularity of the OP chain sits near $n_\mu \gg n_{\text{stab}}$.*

3 Resolution

k_z	N	n_{stab}	n_{res}	$n_{\text{stab}}/n_{\text{res}}$	n_{qbreak}
5	72	13	143	0.091	24
9	184	13	367	0.035	46
17	96	18	191	0.094	19
26	364	46	727	0.063	60
43	839	60	1677	0.036	119
69	5690	119	11379	0.010	$>2n_{\text{stab}}$
116	1433	158	2865	0.055	159
136	1641	175	3281	0.053	176
137	8300	175	16599	0.011	$>2n_{\text{stab}}$
170	5771	194	11541	0.017	$>2n_{\text{stab}}$
197	4071	406	8141	0.050	407
230	2012	194	4023	0.048	195
500 ^f	17847	1525	35693	0.043	1526

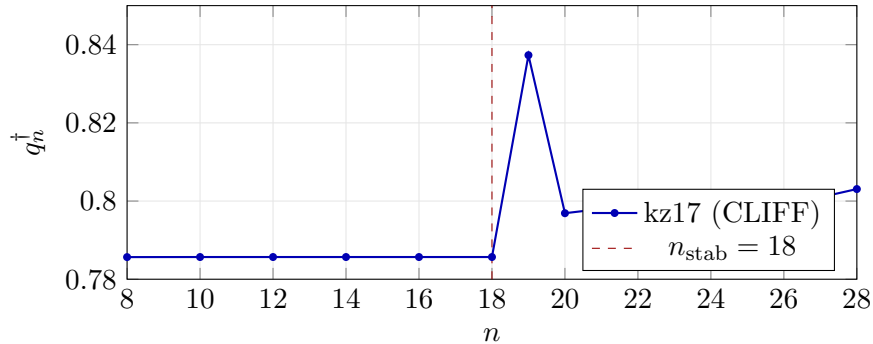
^ffloat Gram; residual $1.5 \cdot 10^{-15}$ at n_{stab} .

Proposition 2 (RES_MISMATCH). *On every measured window, $n_{\text{stab}}/n_{\text{res}} \in [0.010, 0.094]$. The prefix does not resolve individual fold atoms. Hypothesis $n_{\text{stab}} \approx n_{\text{res}}$ is false. The weaker bandlimited statement holds: the extraction prefix sits well below Nyquist.*



4 Schur energy

Proposition 3 (PREFIX_Q_PLATEAU). *On every measured window, including kz137/170/500, $q_n^\dagger$ is constant for $n \leq n_{\text{stab}}$ (plateau std $< 10^{-4}$, Gram residual $\leq 10^{-14}$, $q_* < 1$). The uniform bound $q_n^\dagger \leq q_* < 1$ therefore holds on the r449 prefix. The exit of the plateau is not a universal cliff at n_{stab} : CLIFF at $n_{\text{stab}}+1$ on $\{17, 116, 136, 197, 230, 500\}$; DELAYED on $\{5, 9, 26, 43\}$; CONTINUES past $2n_{\text{stab}}$ on $\{69, 137, 170\}$.*



kz136 (live, $N = 1641$): $q_{175} = 0.87092$, $q_{176} = 0.87551$ (CLIFF). kz137 (dead tail, $N = 8300$, same $n_{\text{stab}} = 175$): q stays 0.90011 through $n = 369$ (CONTINUES). Same cut, different exit — the cliff is not the definition of n_{stab} .

5 Scramble

Proposition 4 (Construction-specific plateau). *Theta-jitter of kz17 (seed 451): $n_{\text{stab}} \rightarrow 2$, n_{res} blows up to $9 \cdot 10^5$, $q_n^\dagger$ wanders (std 0.09, max > 1). The live plateau is fold-arithmetic, not a generic OP artefact of 138 atoms. The live n_{res} -mismatch survives (scramble still has $n_{\text{stab}} \ll n_{\text{res}}$): Nyquist is support geometry; n_{stab} and the plateau are the window comparison.*

6 Consequence for the theorem candidate

n_{stab} is a *measurement cut* (neighbour-moment disagreement), not a unique mechanical cliff of conditioning or of atom resolution. It is nevertheless a *sufficient* depth: on $n \leq n_{\text{stab}}$ the compressed $R^\dagger$ is positivity-compatible ($q_n^\dagger \equiv q_* < 1$) and bandlimited ($n \ll n_{\text{res}}$). The Prefix Stability Theorem can name that plateau; it cannot claim that instability begins exactly at the cut.

Claim boundary

Nothing here is evidence for or against the Riemann hypothesis, in either direction. The identities are a finite moment/chain/pack census.

References

- [1] Flip versus stabilization; n_{stab} .
- [2] Slice floor; $q^\dagger$ pack.
- [3] Prefix mincut; n_{stab} -compression.